Here's an updated list of **Java Developer Skills for 2025**, aligned with current trends, tools, and industry demands:

🚀 Core Java Skills (Mandatory)

1. **Java 17+ or Java 21 (LTS)**
   - Modern features: Records, Sealed Classes, Pattern Matching, Switch Expressions, etc.
2. **OOPs Concepts & Design Patterns**
   - SOLID principles, Factory, Singleton, Strategy, Observer, etc.
3. **Multithreading & Concurrency**
   - java.util.concurrent, CompletableFuture, parallel streams.
4. **Collections Framework & Streams API**
5. **Exception Handling, I/O, and File Handling (NIO)**

🧱 Frameworks & Libraries

1. **Spring Ecosystem** (latest versions)
   - **Spring Boot 3** (native support for GraalVM)
   - Spring MVC, Spring Data JPA, Spring Security, Spring Batch
   - Spring WebFlux (Reactive Programming)
2. **Hibernate / JPA**
3. **Maven / Gradle** (Build Tools)
4. **Lombok**, **MapStruct**, **ModelMapper**

🌐 Web & API Development

1. **RESTful API Design**
2. **JSON/XML Parsing**
3. **Postman / Swagger / OpenAPI**
4. **GraphQL** (optional but useful)

🗄️ Database & Caching

1. **SQL** – MySQL, PostgreSQL
2. **NoSQL** – MongoDB, Cassandra, Redis
3. **JDBC / JPA / QueryDSL**

☁️ DevOps & Cloud (High Demand in 2025)

1. **Docker & Kubernetes**
2. **CI/CD Pipelines** – Jenkins, GitHub Actions, GitLab CI
3. **Cloud Platforms** – AWS (EC2, RDS, S3, Lambda), GCP, Azure
4. **Terraform / Helm** (bonus)

🎛️ Frontend & Full Stack (Optional, if Full Stack)

1. **HTML5, CSS3, JavaScript**
2. **Angular / React / Vue.js**
3. **Thymeleaf / JSP (for traditional server-side rendering)**

🧪 Testing

1. **JUnit 5 / TestNG**
2. **Mockito / AssertJ**
3. **Integration Testing** – Spring Boot Test, REST Assured
4. **Performance Testing** – JMeter, Gatling (optional)

🔐 Security & Best Practices

1. **JWT / OAuth2 / Spring Security**
2. **OWASP Top 10 Awareness**
3. **Code Quality** – SonarQube, PMD, Checkstyle

🧠 Tools & Others

1. **Git / GitHub / GitLab / Bitbucket**
2. **IDE** – IntelliJ IDEA (preferred), Eclipse, VS Code
3. **Agile / Scrum Methodologies**
4. **Unit + Code Coverage** – JaCoCo, Cobertura

Top Skills for Java Developer at Google (2025)

1. Strong Computer Science Fundamentals

Google emphasizes deep knowledge in CS theory.

- Data Structures & Algorithms (Arrays, Trees, Graphs, HashMaps)
- Big-O analysis, recursion, sorting/searching
- System Design (scalable, fault-tolerant systems)
- Concurrency & multithreading
- Operating systems & networking basics

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2. Proficiency in Java Programming

Expect to work on large-scale backend systems.

- Java 17+ / Java 21 (LTS)
- OOP Principles & SOLID design
- Generics, Enums, Streams, Lambda
- Exception handling & memory management
- Java Concurrency (java.util.concurrent, thread pools)
- JVM internals (Garbage Collection, tuning is a plus)

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3. Backend Development & API Design

You should know how to build and scale APIs/services.

- Spring Boot / Spring Framework
- RESTful API Design

- Microservices architecture
- gRPC or GraphQL (good to know)
- JSON, XML parsing
- WebFlux or reactive programming (bonus)

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4. Database & Storage Systems

Efficient data storage & retrieval is crucial.

- Relational DBs: MySQL, PostgreSQL
- ORM: Hibernate / JPA
- NoSQL: Bigtable, MongoDB, Redis
- Query optimization and indexing

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5. DevOps, Cloud & Tooling

Google engineers are expected to be full-cycle developers.

- Google Cloud Platform (GCP): App Engine, Compute Engine, Cloud Storage, BigQuery
- Docker & Kubernetes
- Git, CI/CD (Jenkins, GitHub Actions)
- Monitoring: Stackdriver, Prometheus, Grafana

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6. Testing & Quality Assurance

High-quality code is a must at Google.

- Unit Testing: JUnit 5, Mockito
- Integration Testing
- Test-driven development (TDD)
- Static code analysis: SonarQube

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7. Version Control & Development Tools

- Git (command-line proficiency)

- Build tools: Maven or Gradle
- IntelliJ IDEA (preferred), VS Code, or Eclipse

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8. Soft Skills & Google Culture Fit

Google highly values teamwork, curiosity, and ownership.

- Clear communication (technical & non-technical)
- Collaboration (code reviews, design discussions)
- Problem-solving mindset
- Growth mindset & learning agility
- Familiarity with Agile / Scrum

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9. Optional/Bonus Skills

These can make your profile stand out at Google.

- Distributed systems & CAP theorem
- Kafka or Pub/Sub (messaging systems)
- Machine learning fundamentals (TensorFlow basics)
- OpenTelemetry, tracing, or observability tools
- Secure coding practices (OWASP, OAuth2, JWT)